Taipei Veterans General Hospital

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Sample Information

Patient Name: 林家祥
Gender: Male
ID No.: A110515279
History No.: 32696643
Age: 65

Ordering Doctor: DOC1322F 趙毅
Ordering REQ.: H3LMM45
Signing in Date: 2021/12/03

Path No.: S110-94702
MP No.: F21100
Assay: Oncomine Focus Assay
Sample Type: FFPE
Block No.: S110-77291B
Percentage of tumor cells: 80%

Reporting Doctor: DOC5466K 葉奕成 (Phone: 8#5466)

Note:

Sample Cancer Type: Cholangiocarcinoma

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Report Highlights

1 Relevant Biomarkers
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1 Clinical Trials

Relevant Biomarkers

Tier	Genomic Alteration	Relevant Therapies (In this cancer type)	Relevant Therapies (In other cancer type)	Clinical Trials
IA	*IDH1 p.(R132H) c.395G>A* isocitrate dehydrogenase (NADP(+)) 1 Allele Frequency: 23.33% **Prognostic significance:** NCCN: Poor **Diagnostic significance:** None	ivosidenib	None	1

Public data sources included in relevant therapies: FDA[1], NCCN, **EMA**[2], ESMO
Public data sources included in prognostic and diagnostic significance: NCCN, ESMO
Tier Reference: Li et al. *Standards and Guidelines for the Interpretation and Reporting of Sequence Variants in Cancer: A Joint Consensus Recommendation of the Association for Molecular Pathology, American Society of Clinical Oncology, and College of American Pathologists.* J Mol Diagn. 2017 Jan;19(1):4-23.

Disclaimer: The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. The data version is 2021.09(004).The content of this report has not been evaluated or approved by the FDA or other regulatory agencies.

Variant Details

DNA Sequence Variants

Gene	Amino Acid Change	Coding	Variant ID	Locus	Allele Frequency	Transcript	Variant Effect	Coverage
IDH1	p.(R132H)	c.395G>A	COSM28746	chr2:209113112	23.33%	NM_005896.3	missense	1997
JAK1	p.(P733=)	c.2199A>G	.	chr1:65310489	33.35%	NM_002227.4	synonymous	1997
PDGFRA	p.(G313=)	c.939T>G	.	chr4:55133726	63.85%	NM_006206.6	synonymous	1997
FGFR4	p.(P136L)	c.407C>T	.	chr5:176517797	99.30%	NM_213647.3	missense	1999
EGFR	p.(Q787=)	c.2361G>A	.	chr7:55249063	51.30%	NM_005228.5	synonymous	2000
RET	p.(L769=)	c.2307G>T	.	chr10:43613843	100.00%	NM_020975.6	synonymous	1997
RET	p.(S904=)	c.2712C>G	.	chr10:43615633	52.50%	NM_020975.6	synonymous	1998

Biomarker Descriptions

IDH1 (isocitrate dehydrogenase (NADP(+)) 1)

Background: The IDH1 and IDH2 genes encode homologous isocitrate dehydrogenase enzymes that catalyze the conversion of isocitrate to α-ketoglutarate (α-KG)[1]. The IDH1 gene encodes the NADP+ dependent cytoplasmic isocitrate dehydrogenase enzyme; IDH2 encodes the mitochondrial isoform.

Alterations and prevalence: Recurrent somatic mutations in IDH1 and IDH2 are mutually exclusive and observed in several malignancies including glioma, chondrosarcoma, intrahepatic cholangiocarcinoma, acute myeloid leukemia (AML), and myelodysplastic syndrome (MDS)[2]. Recurrent IDH1 variants include predominately R132H/C plus other substitutions at lower frequencies. These gain of function variants confer neomorphic enzyme activity[3]. Although wild-type enzymatic activity is ablated, recurrent IDH1 variants catalyze the conversion of α-KG to D-2-hydroxyglutarate, an oncometabolite with diverse effects on cellular metabolism, epigenetic regulation, redox states, and DNA repair[1,4]. Recurrent IDH1 mutations are present in 5-10% of patients with AML and 5% of patients with MDS[5,6,7]. Recurrent IDH1 mutations are present in nearly 80% of lower grade diffuse gliomas[8,9].

Potential relevance: Ivosidenib[10] is FDA approved (2018) for the treatment of AML patients with IDH1 R132C/G/H/L/S variants[11]. Ivosidenib has also been granted breakthrough designation (2020) for IDH1 mutated relapsed or refractory myelodysplastic syndrome (MDS)[12]. IDH1 mutations are associated with inferior leukemia-free survival in primary myelofibrosis (PMF) and inferior overall survival in polycythemia vera (PV) but have been shown to confer improved prognosis in lower grade gliomas[13,14,15].

Relevant Therapy Summary

● In this cancer type ○ In other cancer type ◑ In this cancer type and other cancer types ✖ No evidence

IDH1 p.(R132H) c.395G>A

Relevant Therapy	FDA	NCCN	EMA	ESMO	Clinical Trials*
ivosidenib	✖	●	✖	✖	✖
LY-3410738	✖	✖	✖	✖	● (I)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

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Relevant Therapy Details

Current NCCN Information

● In this cancer type ○ In other cancer type ◑ In this cancer type and other cancer types

NCCN information is current as of 2021-08-02. For the most up-to-date information, search www.nccn.org.
For NCCN International Adaptations & Translations, search www.nccn.org/global/international_adaptations.aspx.

IDH1 p.(R132H) c.395G>A

● ivosidenib

Cancer type: Extrahepatic Cholangiocarcinoma, Intrahepatic Cholangiocarcinoma

Variant class: IDH1 mutation

NCCN Recommendation category: 2A

Population segment (Line of therapy):
- Unresectable, Metastatic, Progression (Subsequent therapy); Useful in certain circumstances

Reference: NCCN Guidelines® - NCCN-Hepatobiliary Cancers [Version 3.2021]

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Prognostic Details

Current NCCN Information

NCCN information is current as of 2021-08-02. For the most up-to-date information, search www.nccn.org.
For NCCN International Adaptations & Translations, search www.nccn.org/global/international_adaptations.aspx.

IDH1 p.(R132H) c.395G>A

Prognostic significance: NCCN: Poor

Cancer type: Extrahepatic Cholangiocarcinoma **Variant class:** IDH1 mutation

NCCN Recommendation category: 2A

Summary:
- NCCN® associates the biomarker with poor prognosis

Reference: NCCN Guidelines® - NCCN-Hepatobiliary Cancers [Version 3.2021]

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Clinical Trials in Taiwan region:

Clinical Trials Summary

IDH1 p.(R132H) c.395G>A

NCT ID	Title	Phase
NCT04521686	Study of LY3410738 Administered to Patients With Advanced Solid Tumors With IDH1 Mutations.	I

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Signatures

Testing Personnel: Shin-Ying Lin 2021.12.08 10:50:42 +08'00'

Laboratory Supervisor: 2021.12.08 10:50:54 +08'00'

Pathologist: Yi-Chu Yeh 2021.12.08 10:51:21 +08'00'

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References

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